

torch.nonzero#

<https://pytorch.org/docs/stable/generated/torch.nonzero.html>

Description:

`torch.nonzero(..., as_tuple=False)` (default) returns a 2-D tensor where each row is the index for a nonzero value. `torch.nonzero(..., as_tuple=True)` returns a tuple of 1-D index tensors, allowing for advanced indexing, so `x[x.nonzero(as_tuple=True)]` gives all nonzero values of tensor `x`. Of the returned tuple, each index tensor contains nonzero indices for a certain dimension. See below for more details on the two behaviors. When input is on CUDA, `torch.nonzero()` causes host-device synchronization.

Function Signature

```
torch.nonzero(input, *, out=None, as_tuple=False) →  
LongTensor or tuple of LongTensors#
```

Parameters

Keyword Arguments

Returns

Return type

Parameters

Keyword

out (LongTensor, optional) – the output tensor containing indices

Returns:

LongTensor or tuple of LongTensor

Code Examples

```
>>> torch.nonzero(torch.tensor([1, 1, 1, 0, 1]))
tensor([[ 0],
        [ 1],
        [ 2],
        [ 4]])

>>> torch.nonzero(torch.tensor([[0.6, 0.0, 0.0, 0.0],
...                             [0.0, 0.4, 0.0, 0.0],
...                             [0.0, 0.0, 1.2, 0.0],
...                             [0.0, 0.0, 0.0, -0.4]]))
tensor([[ 0,  0],
        [ 1,  1],
        [ 2,  2],
        [ 3,  3]])

>>> torch.nonzero(torch.tensor([1, 1, 1, 0, 1]), as_tuple=True)
(tensor([0, 1, 2, 4]),)

>>> torch.nonzero(torch.tensor([[0.6, 0.0, 0.0, 0.0],
...                             [0.0, 0.4, 0.0, 0.0],
...                             [0.0, 0.0, 1.2, 0.0],
...                             [0.0, 0.0, 0.0, -0.4]]),
as_tuple=True)
(tensor([0, 1, 2, 3]), tensor([0, 1, 2, 3]))

>>> torch.nonzero(torch.tensor(5), as_tuple=True)
(tensor([0]),)
```

Notes & Warnings

NOTE

Note `torch.nonzero(..., as_tuple=False)` (default) returns a 2-D tensor where each row is the index for a nonzero value. `torch.nonzero(..., as_tuple=True)` returns a tuple of 1-D index tensors, allowing for advanced indexing, so `x[x.nonzero(as_tuple=True)]` gives all nonzero values of tensor `x`. Of the returned tuple, each index tensor contains nonzero indices for a certain dimension. See below for more details on the two behaviors. When input is on CUDA, `torch.nonzero()` causes host-device synchronization.

Detailed Documentation

Documentation

`torch.nonzero(..., as_tuple=False)` (default) returns a 2-D tensor where each row is the index for a nonzero value.

`torch.nonzero(..., as_tuple=True)` returns a tuple of 1-D index tensors, allowing for advanced indexing, so `x[x.nonzero(as_tuple=True)]` gives all nonzero values of tensor `x`. Of the returned tuple, each index tensor contains nonzero indices for a certain dimension.

See below for more details on the two behaviors.

When input is on CUDA, `torch.nonzero()` causes host-device synchronization.

When `as_tuple` is `False` (default):

Returns a tensor containing the indices of all non-zero elements of input. Each row in the result contains the indices of a non-zero element in input. The result is sorted lexicographically, with the last index changing the fastest (C-style).

If input has n dimensions, then the resulting indices tensor out is of size $(z \times n) \times n$, where z is the total number of non-zero elements in the input tensor.

When `as_tuple` is `True`:

Returns a tuple of 1-D tensors, one for each dimension in input, each containing the indices (in that dimension) of all non-zero elements of input .

If input has n dimensions, then the resulting tuple contains n tensors of size z , where z is the total number of non-zero elements in the input tensor.

As a special case, when input has zero dimensions and a nonzero scalar value, it is treated as a one-dimensional tensor with one element.

input (Tensor) – the input tensor.

out (LongTensor, optional) – the output tensor containing indices

If `as_tuple` is `False`, the output tensor containing indices. If `as_tuple` is `True`, one 1-D tensor for each dimension, containing the indices of each nonzero element along that dimension.

LongTensor or tuple of LongTensor